GCSE PHYSICS

Higher Tier – Paper 2 style

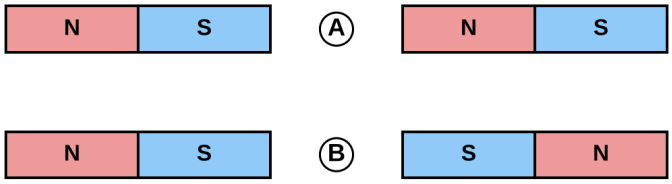
Topic 4.7: Magnetism and electromagnetism

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 130 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **125**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 125

0 1 The diagram shows two pairs of bar magnets, **A** and **B**. In each pair, the magnets are placed with a small gap between them.



0 1 . 1 Describe what happens in pair **A** and in pair **B**. **[2 marks]**

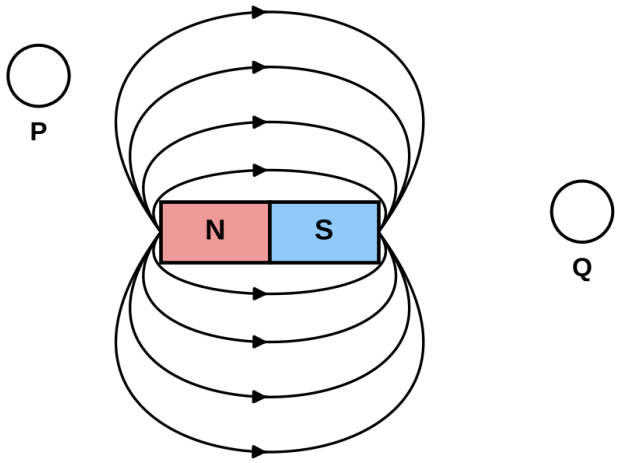
0 1 . 2 State the rule for forces between magnetic poles. **[2 marks]**

0 1 . 3 Name **three** magnetic materials. **[3 marks]**

0 1 . 4 Explain the difference between a permanent magnet and an induced magnet. **[3 marks]**

0 1 . 5 Explain why an unmagnetised iron nail is attracted to both ends of a bar magnet. **[2 marks]**

0 2 The diagram shows a bar magnet. The magnetic field lines are shown. **P** and **Q** are plotting compasses.



0 2 . 1 Draw an arrow inside each compass to show the direction the needle points. **[2 marks]**

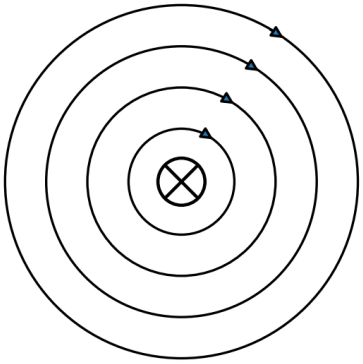
0 2 . 2 State what is meant by a **magnetic field**. **[1 mark]**

0 2 . 3 State where the magnetic field is strongest. Give a reason from the diagram. **[2 marks]**

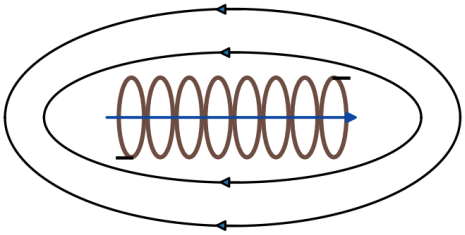
0 2 . 4 Describe how a plotting compass can be used to plot the magnetic field around a bar magnet. **[3 marks]**

0 2 . 5 Explain why a compass needle points north when it is far from any magnet. **[3 marks]**

0 3 The diagrams show the magnetic field around a straight wire and around a solenoid when a current flows.



current into the page



solenoid

0 3 . 1 Describe the shape of the magnetic field around the straight wire carrying a current. **[2 marks]**

0 3 . 2 The current in the wire is into the page. State the direction of the field lines. **[1 mark]**

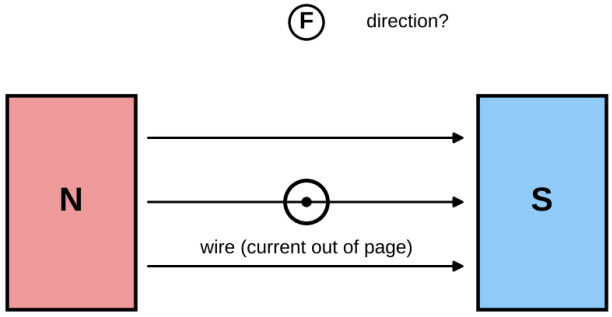
0 3 . 3 State **two** ways of increasing the strength of the magnetic field around the wire. **[2 marks]**

0 3 . 4 Describe the magnetic field inside and outside a solenoid. **[3 marks]**

0 3 . 5 State **three** ways of increasing the strength of the field of a solenoid. **[3 marks]**

0 3 . 6 Give **one** use of an electromagnet. **[1 mark]**

0 4 A wire carrying a current is placed in a magnetic field between two magnets, as shown. The current is out of the page.



0 4 . 1 State the direction of the force on the wire. **[1 mark]**

0 4 . 2 Describe how the direction of the force on the wire can be reversed. **[2 marks]**

0 4 . 3 Write down the equation that links force, magnetic flux density, current and length. **[1 mark]**

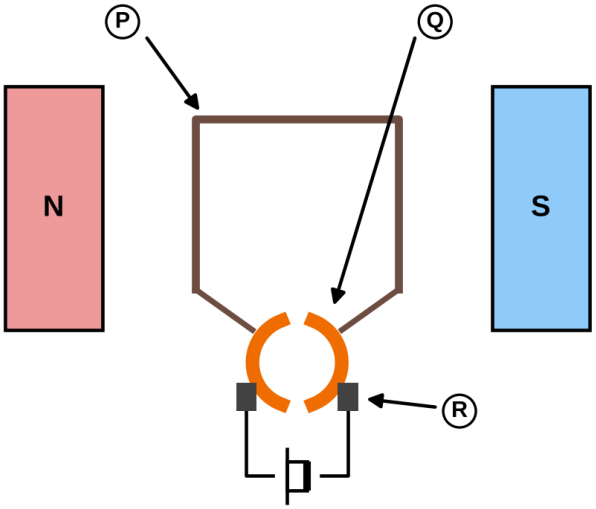
0 4 . 4 The magnetic flux density is 0.30 T, the current is 5.0 A and the length of wire in the field is 0.12 m. Calculate the force on the wire. **[3 marks]**

Answer = _____ N

0 4 . 5 State the force on the wire if it is placed parallel to the magnetic field. **[1 mark]**

0 4 . 6 Name **two** things that must be at right angles to each other for the motor effect to occur. **[2 marks]**

0 5 The diagram shows a simple d.c. motor.



0 5 . 1 Name the parts labelled **P**, **Q** and **R**. **[3 marks]**

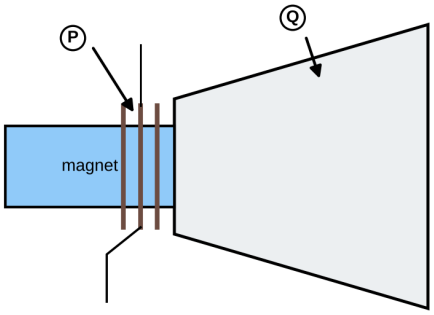
0 5 . 2 Explain how the motor makes the coil turn. **[4 marks]**

0 5 . 3 Explain the function of the split-ring commutator. **[2 marks]**

0 5 . 4 Give **two** ways of increasing the speed of the motor. **[2 marks]**

0 5 . 5 State how to make the motor turn the other way. **[1 mark]**

0 6 The diagram shows part of a loudspeaker.



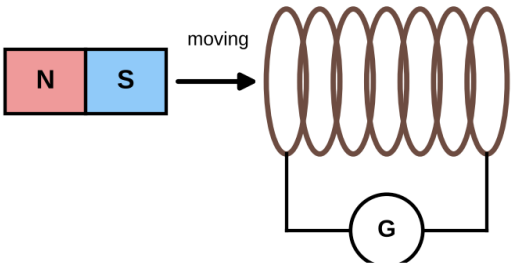
0 6 . 1 Name the parts labelled **P** and **Q**. **[2 marks]**

0 6 . 2 Explain how the loudspeaker produces sound. **[4 marks]**

0 6 . 3 State what determines the frequency of the sound produced. **[1 mark]**

0 6 . 4 A louder sound is needed. Describe the change to the electrical signal. **[1 mark]**

0 7 A student moves a bar magnet into a coil of wire connected to a sensitive voltmeter (galvanometer).



0 7 . 1 Describe what the student observes when the magnet is pushed into the coil. **[1 mark]**

0 7 . 2 State what happens when the magnet is held still inside the coil. **[1 mark]**

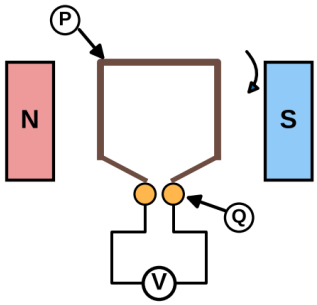
0 7 . 3 Give **three** ways of increasing the size of the induced p.d. **[3 marks]**

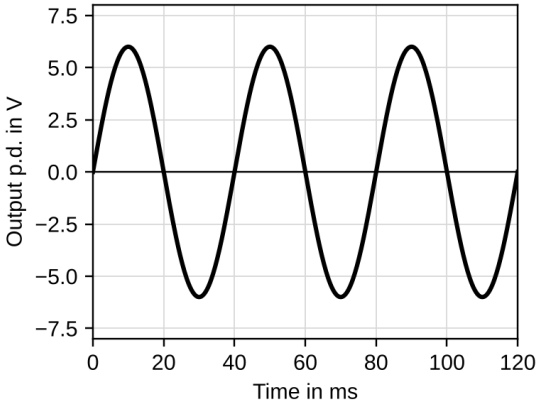
0 7 . 4 Describe what happens to the meter when the magnet is pulled out of the coil. **[2 marks]**

0 7 . 5 Explain why a p.d. is induced. **[2 marks]**

0 7 . 6 The induced current creates its own magnetic field that opposes the movement of the magnet. Explain why it is harder to push the magnet into the coil when the circuit is complete. **[2 marks]**

0 8 The diagram shows an a.c. generator (alternator) and a graph of its output.





0 8 . 1 Name the parts labelled **P** and **Q**. **[2 marks]**

0 8 . 2 Explain how the generator produces an alternating p.d. **[4 marks]**

0 8 . 3 Use the graph to find the time for one complete cycle and calculate the frequency of the output. **[4 marks]**

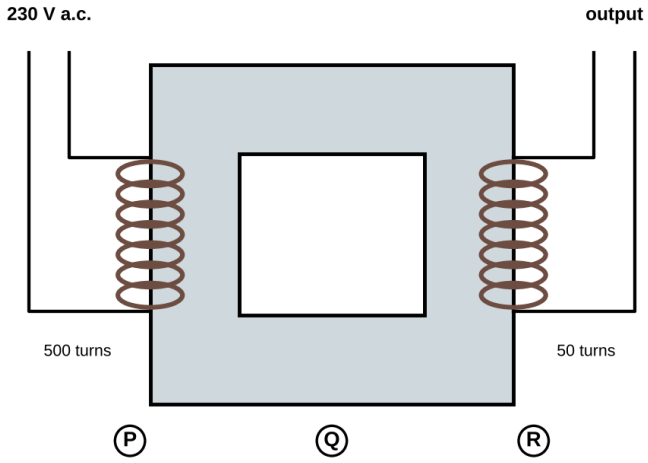
Answer = _____ Hz

0 8 . 4 The coil is rotated at twice the speed. Describe **two** changes to the graph. **[2 marks]**

0 8 . 5 A dynamo produces direct current instead. State what part replaces the slip rings. **[1 mark]**

0 8 . 6 Explain how a microphone converts sound into an electrical signal. **[3 marks]**

0 9 The diagram shows a transformer.



0 9 . 1 Name the parts labelled **P**, **Q** and **R**. **[3 marks]**

0 9 . 2 Explain how a transformer works. **[4 marks]**

0 9 . 3 Explain why a transformer does not work with direct current. **[2 marks]**

0 9 . 4 Write down the equation that links the p.d.s across the coils to the number of turns. **[1 mark]**

0 9 . 5 The primary coil has 500 turns and is connected to 230 V. The secondary coil has 50 turns. Calculate the p.d. across the secondary coil. **[3 marks]**

Answer = _____ V

0 9 . 6 The transformer is 100% efficient. The secondary current is 4.0 A. Calculate the current in the primary coil. **[4 marks]**

Answer = _____ A

0 9 . 7 A transformer is needed to supply 12 V from the 230 V mains using a primary coil of 1000 turns. Calculate the number of turns on the secondary coil. **[3 marks]**

Answer = _____ turns

0 9 . 8 State whether the transformer in part 5 is a step-up or a step-down transformer. **[1 mark]**

1 0 Read the question and answer in full sentences.

1 0 . 1 *Explain how a d.c. motor works. Your answer should include the motor effect, the role of the split-ring commutator, and one way of changing the speed. **[6 marks]**

1 0 . 2 *Explain why the National Grid uses transformers, and describe what happens to the p.d. and current at each stage between a power station and homes. Use the equation P = V I in your answer. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.7: Magnetism and electromagnetism

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	A: the magnets attract ; B: the magnets repel ;		2	AO2 4.7.1.1
01.2	like poles repel ; unlike poles attract ;		2	AO1 4.7.1.1
01.3	iron ; steel ; cobalt ; nickel ;	Any three.	3	AO1 4.7.1.1
01.4	permanent magnet produces its own magnetic field ; induced magnet becomes magnetic only when in a magnetic field ; induced magnet loses most / all magnetism when removed from the field ;		3	AO1 4.7.1.1
01.5	the nail becomes an induced magnet ; poles of the nail are opposite to those of the magnet nearest it, so it attracts ;		2	AO2 4.7.1.1
02.1	P: along the tangent to the field line, pointing away from N / towards S ; Q: pointing towards S (along the field line direction) ;		2	AO2 4.7.1.2
02.2	region where a magnet / magnetic material experiences a force		1	AO1 4.7.1.2
02.3	at the poles ; field lines are closest together ;		2	AO2 4.7.1.2
02.4	place the compass near one pole and mark the position of the needle ends ; move the compass so the tail is where the head was and repeat ; join the marks to draw a field line (and show the direction N to S) ;		3	AO1 4.7.1.2
02.5	the Earth has a magnetic field ; the core of the Earth is magnetic (iron) ; the compass needle is a small magnet that aligns with the field ;		3	AO2 4.7.1.2
03.1	concentric circles ; around the wire (in a plane at right angles to the wire) ;		2	AO1 4.7.2.1
03.2	clockwise	Use the right-hand grip rule.	1	AO2 4.7.2.1
03.3	increase the current ; move closer to the wire ;		2	AO2 4.7.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	inside: strong and uniform (parallel field lines) ; outside: similar to the field of a bar magnet ; field lines come out one end and re-enter the other end ;		3	AO1 4.7.2.2
03.5	increase the current ; increase the number of turns (coils) ; add a soft iron core ;		3	AO1 4.7.2.2
03.6	scrapyard crane / relay / electric bell / lock / MRI scanner / loudspeaker		1	AO1 4.7.2.2
04.1	upwards / up the page	Use Fleming's left-hand rule.	1	AO2 4.7.2.3
04.2	reverse the current ; or reverse the magnetic field (swap the magnets) ;		2	AO1 4.7.2.3
04.3	force = magnetic flux density × current × length		1	AO1 4.7.2.3
04.4	F = B I l ; = 0.30 × 5.0 × 0.12 ; 0.18 (N) ;		3	AO2 4.7.2.3
04.5	zero / no force		1	AO2 4.7.2.3
04.6	current in the wire ; magnetic field ;		2	AO1 4.7.2.3
05.1	P: coil ; Q: split-ring commutator ; R: brush ;		3	AO1 4.7.2.4
05.2	current in the coil in a magnetic field gives a force (motor effect) ; the current is in opposite directions in the two sides of the coil ; forces are in opposite directions on the two sides ; giving a turning effect (moment) ;		4	AO2 4.7.2.4
05.3	it reverses the current in the coil every half turn ; so the coil keeps turning in the same direction ;		2	AO2 4.7.2.4
05.4	increase the current / p.d. ; use stronger magnets ; more turns on the coil ;	Any two.	2	AO1 4.7.2.4
05.5	reverse the current / swap the battery or the magnets		1	AO1 4.7.2.4
06.1	P: coil ; Q: (paper) cone ;		2	AO1 4.7.2.5
06.2	alternating current in the coil ; produces a changing magnetic field which interacts with the permanent magnet ; causing a force that makes the coil and cone move backwards and forwards ; making the air vibrate (sound wave) ;		4	AO2 4.7.2.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	the frequency of the alternating current		1	AO2 4.7.2.5
06.4	increase the amplitude / current of the signal		1	AO2 4.7.2.5
07.1	the meter deflects briefly (a p.d. is induced)		1	AO1 4.7.3.1
07.2	no deflection / no p.d. induced		1	AO1 4.7.3.1
07.3	move the magnet faster ; use a stronger magnet ; more turns on the coil ;		3	AO1 4.7.3.1
07.4	the meter deflects in the opposite direction ; p.d. is in the opposite direction ;		2	AO2 4.7.3.1
07.5	the wire (coil) cuts through magnetic field lines / the magnetic field through the coil changes ; this is the generator effect ;		2	AO2 4.7.3.1
07.6	work has to be done against the opposing force ; this energy is transferred to the electrical energy of the current ;		2	AO3 4.7.3.1
08.1	P: coil ; Q: slip ring ;		2	AO1 4.7.3.1
08.2	the coil rotates in the magnetic field ; the wires cut field lines, inducing a p.d. ; the direction of the p.d. reverses each half turn ; slip rings and brushes connect the coil to the external circuit ;		4	AO2 4.7.3.2
08.3	period = 40 ms ; = 0.040 s ; f = 1 ÷ T ; 25 (Hz) ;		4	AO3 4.7.3.2
08.4	peak p.d. doubles / greater amplitude ; frequency doubles / period halves ;		2	AO3 4.7.3.2
08.5	split-ring commutator		1	AO2 4.7.3.2
08.6	sound waves cause the diaphragm to vibrate ; attached coil moves in a magnetic field ; inducing an alternating p.d. with the same frequency as the sound ;		3	AO2 4.7.3.3
09.1	P: primary coil ; Q: (soft) iron core ; R: secondary coil ;		3	AO1 4.7.3.4
09.2	alternating current in the primary coil ; produces a changing magnetic field (in the iron core) ; which induces an alternating p.d. in the secondary coil ; the number of turns decides whether the p.d. is stepped up or down ;		4	AO2 4.7.3.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.3	d.c. gives a constant magnetic field ; needs a changing field to induce a p.d. ;		2	AO2 4.7.3.4
09.4	p.d. across primary ÷ p.d. across secondary = turns on primary ÷ turns on secondary		1	AO1 4.7.3.4
09.5	$230 \div V_s = 500 \div 50$; $V_s = 230 \div 10$; 23 (V) ;		3	AO2 4.7.3.4
09.6	$V_p I_p = V_s I_s$; $230 \times I_p = 23 \times 4.0$; $I_p = 92 \div 230$; 0.40 (A) ;		4	AO3 4.7.3.4
09.7	$N_s = 1000 \times 12 \div 230$; 52.2 ; 52 (turns) ;		3	AO2 4.7.3.4
09.8	step-down		1	AO2 4.7.3.4
10.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): clear, linked explanation with all three points correct and in sequence. • Level 2 (3–4): some correct explanation of motor effect and commutator. • Level 1 (1–2): simple statements. • Indicative content: current in the coil in a magnetic field; force on each side (Fleming's left-hand rule); sides carry current in opposite directions so forces are opposite; coil rotates; commutator reverses the current every half turn; keeps coil turning in the same direction; speed increased with stronger magnet / more current / more turns.		6	AO2 4.7.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): full explanation of step-up and step-down with p.d., current and power loss reasoning. • Level 2 (3–4): some explanation. • Level 1 (1–2): simple statements. • Indicative content: step-up transformer raises p.d. to e.g. 400 000 V; for the same power, current is lower ($P = V I$); lower current means less power lost as heat ($P = I^2R$); efficient transmission; step-down transformers near homes reduce the p.d. to 230 V for safety; transformers need a.c.		6	AO3 4.7.3.4

Where marks are lost in Topic 4.7

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Field lines** go from N to S outside a magnet. Closer lines = stronger field. Plotting compasses point along the field line towards S.
- ☐ **Right-hand grip rule:** thumb = current direction, fingers = field direction. For an end-on view, current into the page gives a clockwise field.
- ☐ **Fleming's left-hand rule (motor effect):** First finger = Field, seCond finger = Current, thuMb = Motion (force). They are mutually at right angles.
- ☐ **F = B I l:** length is the length of wire **in** the field; no force if the wire is parallel to the field.
- ☐ **Generator effect:** p.d. is induced when there is relative movement between a conductor and a magnetic field. Faster, stronger, more turns = bigger p.d.
- ☐ **Transformer:** $V_p \div V_s = N_p \div N_s$ and $V_p I_p = V_s I_s$. A transformer only works with a.c. The secondary has fewer turns for step-down.
- ☐ **Motor vs generator:** motor – electrical to kinetic (current → force); generator – kinetic to electrical (movement → p.d.).
- ☐ **Explain questions:** link cause → effect in steps (current → field → force → rotation) and name the parts (commutator, slip rings, core).